

Aim:- Develop understanding for package & multithreading.

Introduction to Package: A package in Java is a collection of related classes & interfaces grouped together. It helps in arranging, organizing code, avoiding naming act like folders/directories that store similar classes.

Advantage of Package-

- Avoids class name conflicts
- Provides access protections
- Improves code organization
- Supports reusability.

Types of Package-

1. Built in Package:- Predefined package provided by Java.

Ex Java.lang
Java.util
Java.io

2. User-defined Package:- Package created by the programmer using the

package keyword.

creating a package:- program →

Output

Fibonacci Series

0 1 1 2 3 5 8 13 21 34 55
89.

Expt. No. _____

Date _____

Page No. 52.

```
package series;
public class fseries {
    public void display (int n) {
        System.out.println ("Fibonacci series");
        int a = 0, b = 1, c;
        if (n >= 1)
            System.out.print (a + " ");
        if (n >= 2)
            System.out.print (b + " ");
        for (int i = 3; i <= n; i++) {
            c = a + b;
            System.out.print (c + " ");
            a = b;
            b = c;
            System.out.print ("\n");
        }
    }
}

using package;
```

```
import series.fseries;
class print {
    public static void main (String args[]) {
        fseries obj = new fseries ();
        obj.display (12);
    }
}
```


Introduction of Multithreading :- Multithreading is a process of executing multiple threads simultaneously to perform tasks efficiently. A thread is a light weight process that shares memory with other threads.

Advantages

- Better CPU utilization.
- Faster execution
- Efficient source changing
- Improves performance

Thread life Cycle :-

- New
- Runnable
- Running
- Waiting / blocked
- Terminated

Important thread Method -

start() :- Starts thread

run() :- Contain Code

sleep() :- Access thread

Join() :- waits for another thread.

output

In Thread A = 1

In Thread B = 1

In Thread A = 2

In Thread B = 2

In Thread A = 3

In Thread B = 3

Creating a thread:-

```
class a extends thread {  
    public static void run() {  
        for (int i = 1; i <= 5; i++) {  
            System.out.println("Thread a" + i);  
        }  
    }  
}
```

```
class b extends thread {  
    public void run() {  
        for (int i = 1; i <= 5; i++) {  
            System.out.println("In Thread b" + i);  
        }  
    }  
}
```

```
class MultiThread {  
    public static void main (String [] args) {  
        a obj = new a();  
        b obj = new b();  
        obj.start();  
        obj.start();  
    }  
}
```

Synchronization:- Synchronization, closed to control access of multiple threads to shared resources to avoid inconsistency.

Output

a: 1

8

10

Expt. No. _____

Date _____

Page No. 55.

Program to understand Synchronization:-

```
class table {  
    synchronized void printT(int n) {  
        for (int i = 1; i <= 10; i++) {  
            System.out.println(n * i);  
        }  
    }  
}
```

```
class a extends Thread {  
    table t;  
    a(table t) {  
        this.t = t; }  
    public void run() {  
        t.printT(5);  
    }  
}
```

```
class b extends Thread {  
    table t;  
    b(table t) {  
        this.t = t;  
    }  
}
```

```
public void run() {  
    t.printT(10);  
}
```

```
class synTest {  
    public static void main (String[] args) {  
        table obj = new table();  
        a obj1 = new a(obj);  
        b obj2 = new b(obj);  
        obj1.start();  
        obj2.start();  
    }  
}
```

Teacher's Signature _____

Viva Questions

Q1 What is multithreading in Java?

Multithreading is the process of executing multiple threads simultaneously to perform multiple tasks efficiently.

Q2 What is a thread in Java.

A thread is the smallest unit of execution in a program. It is a lightweight subprocess that runs independently. Java supports multithreading, where multiple threads ~~are~~ can run concurrently to perform different tasks.

Q3 How can threads be created in Java?

Threads in Java can be created in two ways:

(a) By extending the thread class.

(b) By implementing the Runnable interface.

After creating the thread, the start() method is used to begin execution of the thread.

Q4 What is a package in Java?

A package in Java is a collection of related classes, interfaces & sub-packages. Packages are used to organize Java programs & provide a namespace to avoid naming conflict between classes.

- Q5. What are the advantages of package in Java?
- Ans. Packages provide several advantage such as:-
- Organizing classes in a structured manner
 - Avoiding naming conflict
 - Providing access protection
 - Improving code reusability & maintainability
 - Making large applications easier to manage.

Aim:- Develop understanding to do exception handling.

Introduction:- Exception Handling is a mechanism used in programming to handle run time errors (Exceptions) in a controlled way.

It ensures that a program does not crash abruptly and can continue execution or terminate peacefully.

An exception is an event that disrupts the normal flow of a program such as:

- Division by zero
- File not found
- Invalid input
- Array index out of bounds

Basic Components of exception handling:-

1. Try block:- Contains code that may generate an exception.
2. Catch block:- Handles the exception thrown in the try block.
3. Finally block:- Executes regardless of exception occurrence.
4. Throws keyword:- Declares exception that a function may generate.
5. Throw keyword:- Used to identify explicitly throws an exception.

Types of exception:-1. Compile time execution:-

- Deleted during compilation
- Example handling errors.

2. Runtime Exception :- Occurs during program execute

Example - Arithmetic exception, Null point exception.

3. Advantage of exception handling:-

- Maintain normal program flow.
- Provides meaningful error message
- Enhances program maintainability
- Handles unexpected situations efficiently.

P-1 class test 1

```
public static void main (String [] args) {
    Key k;
    int a = 10 / 0;
    catch (ArithmeticException e) {
        System.out.println("Exception handled" + e);
    }
}
```

P-2 Class myException extends Exception {

```
myException (String[] args) {
    super(msg); } }
```

Teacher's Signature _____

Output P1

Exception handled

Finally block always
executes.

output

Handled Arithmetic exception
Finally block always execute
Custom exception: Age less than 18

Expt. No. _____

Date _____

Page No. 60

```
class AddMoreException {  
    static void checkAge (int age) throws  
        myException {  
        if (age < 18) {  
            throw new myException ("Age class is Age is  
                less than 18");  
        } else {  
            System.out.println ("Enable for voting");  
        }  
    }  
    public static void main (String [] args) {  
        Key k;  
        Key k;  
        int a = 10, b = 0;  
        int c = a/b;  
        System.out.println (C);  
        catch (ArithmeticException e) {  
            System.out.println ("Handled Arithmetic  
                exception");  
        }  
        int arr[] = new int [3];  
        arr[5] = 10;  
        catch (ArrayIndexOutOfBoundsException e) {  
            System.out.println ("Handled Array index");  
        }  
        finally {  
            System.out.println ("finally block always executes");  
        }  
        Key k;  
        checkAge (15);  
        catch (myException e) {  
            System.out.println ("Custom exception: " + e.getMessage());  
        }  
    }  
}
```


Viva Question:

Q1. What is exception handling in Java?

Exception handling is a mechanism in Java used to handle runtime errors to maintain the normal flow of the program.

Q2. What are checked & unchecked exceptions?

- Checked exceptions are checked at compile time & must be handled.
- Unchecked exceptions occur at runtime & are not checked by the compiler.

Q3. What is the use of try, catch & finally?

- Try block contains the code which may make error or ~~mistake~~ problem.
- Catch block is used to handle exception.
- Finally block is executed & is used for cleanup code.

Q4. What is the difference between throw & throws?

- Throw is used to explicitly ~~show~~ throw an exception.
- Throws is used to declare exceptions in a method.

Q5 Why the exception handling important?
Ans Exception handling prevents program crashes
& helps in handling runtime errors smoothly.